Version 7.4.8

PVsyst - Simulation report

Grid-Connected System

Project: HW1_RE_810601219_scenario2

Variant: New simulation variant

Sheds on ground

System power: 9.00 kWp

Tehran - Iran

Author

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Project summary

Geographical Site
Tehran
Iran

Situation

Latitude	35.69 °N
Longitude	51.42 °E
Altitude	1178 m
Time zone	UTC+3.5

Project settings

Albedo	0.20

Weather data
Tehran
Meteonorm 8.1 (1988-2000), Sat=19% - Synthetic

System summary

Grid-Connected System

Sheds on ground

PV Field Orientation
Fixed plane
Tilt/Azimuth 35 / 0 °

Near Shadings
Linear shadings : Fast (table)

User's needs
Unlimited load (grid)

System information

PV Array

Nb. of modules	20 units
Pnom total	9.00 kWp

Inverters

Nb. of units	0.5 unit
Pnom total	15.00 kWac
Pnom ratio	0.600

Results summary

Produced Energy	15832.15 kWh/year	Specific production	1759 kWh/kWp/year	Perf. Ratio PR	83.11 %

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General parameters

Grid-Connected System

Sheds on ground

PV Field Orientation

Orientation	
Fixed plane	
Tilt/Azimuth	35 / 0 °

Sheds configuration	
Nb. of sheds	6 units
Sizes	
Sheds spacing	2.92 m
Collector width	1.58 m
Ground Cov. Ratio (GCR)	54.2 %
Top inactive band	0.02 m
Bottom inactive band	0.02 m
Shading limit angle	
Limit profile angle	29.8 °

Models used	
Transposition	Perez
Diffuse	Perez, Meteonorm
Circumsolar	separate

Horizon
Free Horizon

Near Shadings
Linear shadings : Fast (table)

User's needs
Unlimited load (grid)

PV Array Characteristics

PV module

Manufacturer	VSUN
Model	VSUN-450-144-MH
(Original PVsyst database)	
Unit Nom. Power	450 Wp
Number of PV modules	20 units
Nominal (STC)	9.00 kWp
Modules	2 string x 10 In series
At operating cond. (50°C)	
Pmpp	8.16 kWp
U mpp	373 V
I mpp	22 A

Inverter

Manufacturer	Huawei Technologies
Model	SUN2000-30KTL-M3-380V
(Original PVsyst database)	
Unit Nom. Power	30.0 kWac
Number of inverters	2 * MPPT 25% 0.5 unit
Total power	15.0 kWac
Operating voltage	200-1000 V
Max. power (=>55°C)	33.0 kWac
Pnom ratio (DC:AC)	0.60

Total PV power

Nominal (STC)	9 kWp
Total	20 modules
Module area	44.2 m²

Total inverter power

Total power	15 kWac
Nb. of inverters	1 unit
	0.5 unused
Pnom ratio	0.60

Array losses

Thermal Loss factor

Module temperature according to irradiance	
Uc (const)	20.0 W/m²K
Uv (wind)	0.0 W/m²K/m/s

DC wiring losses

Global array res.	284 mΩ
Loss Fraction	1.5 % at STC

Module Quality Loss

Loss Fraction	-0.8 %

Module mismatch losses

Loss Fraction	2.0 % at MPP

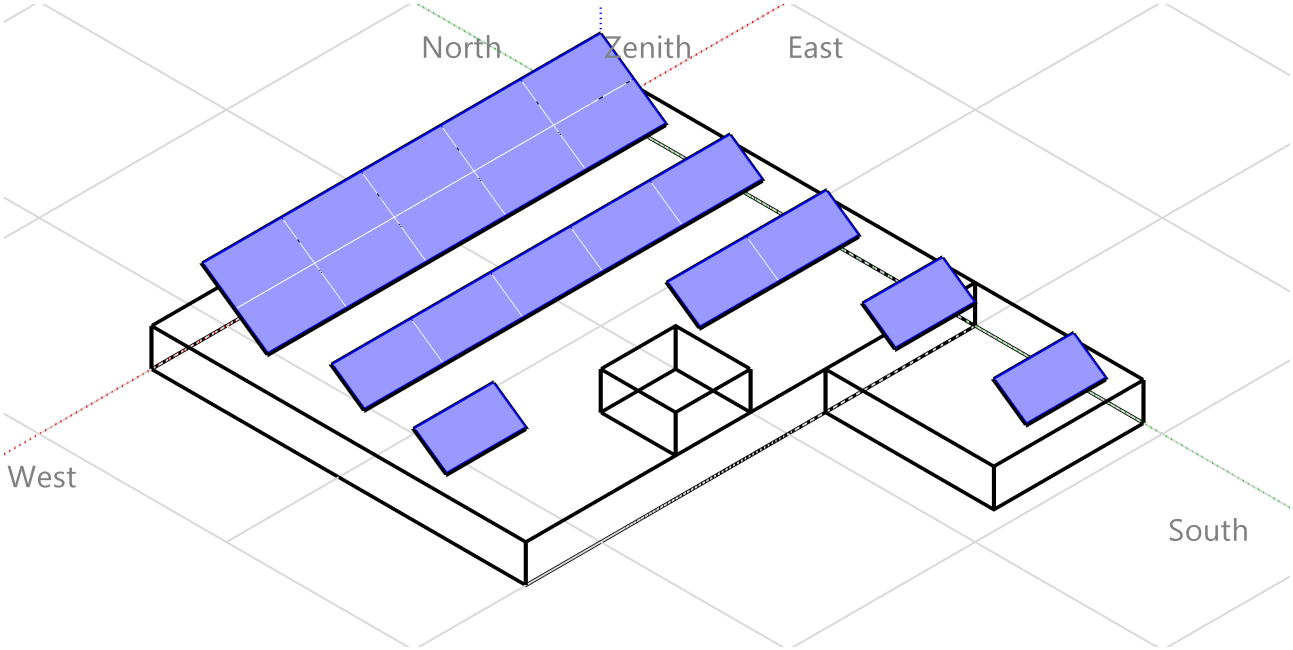
IAM loss factor

Incidence effect (IAM): Fresnel smooth glass, n = 1.526

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.998	0.981	0.948	0.862	0.776	0.636	0.403	0.000

Near shadings parameter

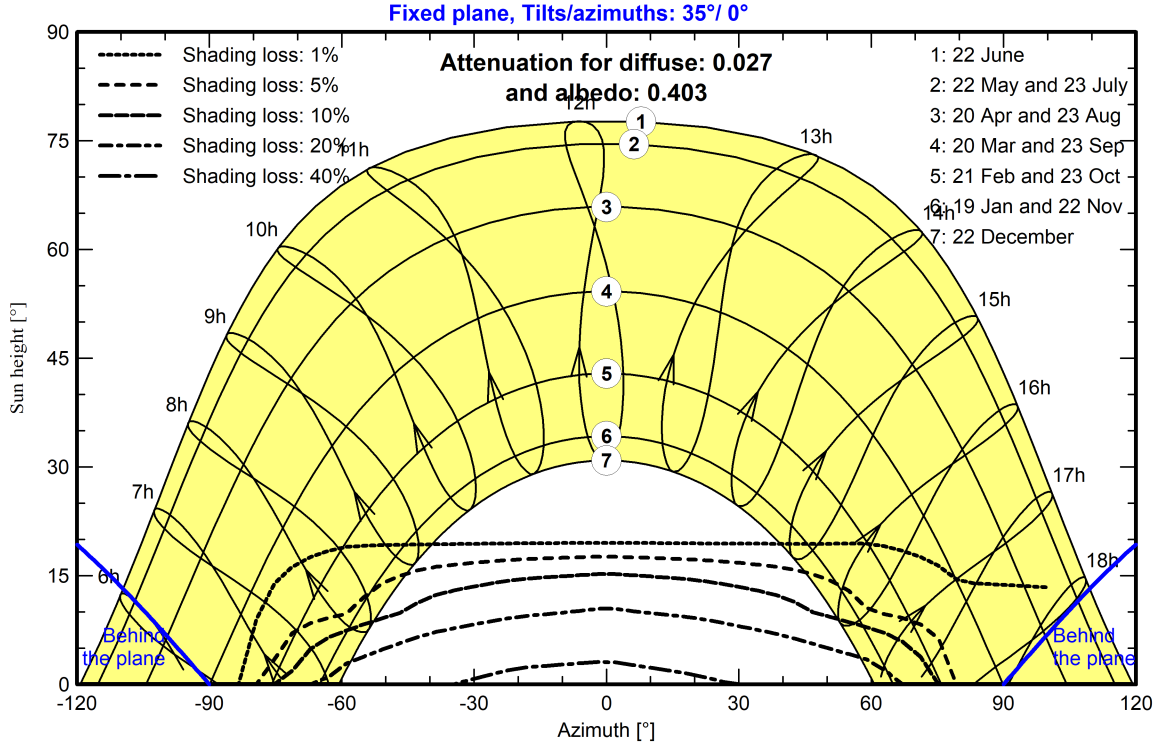
Perspective of the PV-field and surrounding shading scene



Iso-shadings diagram

Orientation #1

Fixed plane, Tilts/azimuths: 35°/ 0°



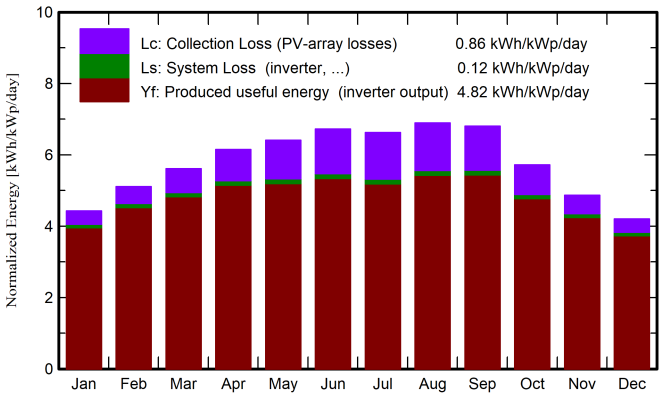
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Main results

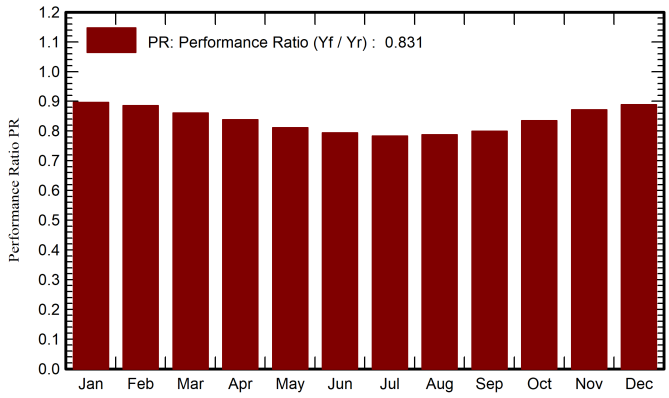
System Production

Produced Energy	15832.15 kWh/year	Specific production	1759 kWh/kWp/year
		Perf. Ratio PR	83.11 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

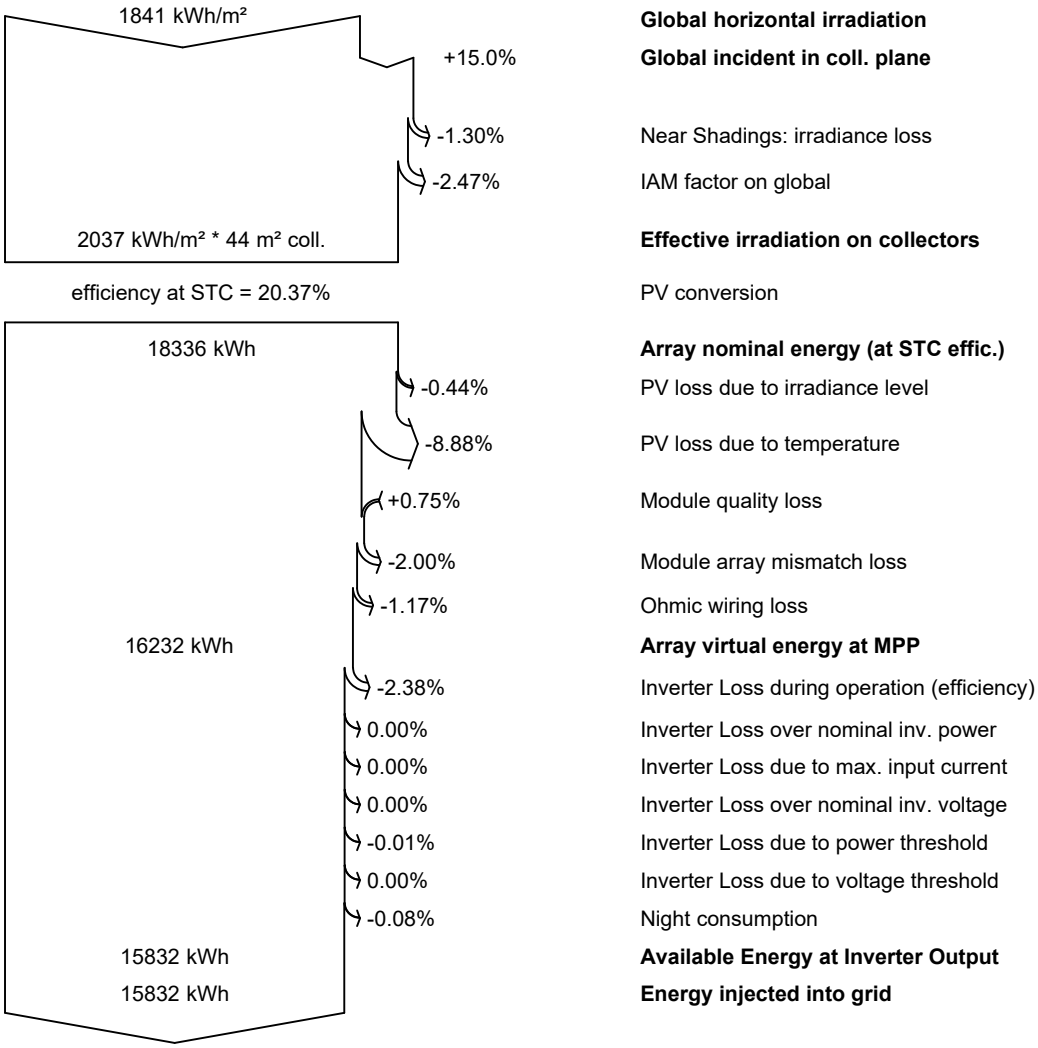
	GlobHor kWh/m²	DiffHor kWh/m²	T_Amb °C	GlobInc kWh/m²	GlobEff kWh/m²	EArray kWh	E_Grid kWh	PR ratio
January	83.3	29.83	5.08	137.1	132.4	1134	1106	0.896
February	99.6	34.42	7.26	143.1	138.8	1169	1141	0.886
March	142.5	54.73	12.75	174.0	167.9	1381	1348	0.860
April	174.2	65.79	17.59	184.6	177.2	1427	1392	0.838
May	209.5	70.09	23.73	198.7	190.1	1489	1451	0.812
June	224.7	73.09	28.75	201.8	192.6	1478	1441	0.793
July	222.3	77.43	31.66	205.3	196.3	1485	1448	0.783
August	209.7	65.80	30.66	213.8	205.4	1553	1515	0.787
September	174.8	45.45	26.24	204.3	197.0	1506	1470	0.800
October	131.8	43.74	19.97	177.4	172.2	1365	1333	0.835
November	91.8	31.48	11.30	146.1	141.1	1175	1146	0.872
December	76.5	26.05	6.44	130.4	126.3	1070	1043	0.888
Year	1840.6	617.91	18.52	2116.6	2037.4	16232	15832	0.831

Legends

GlobHor	Global horizontal irradiation
DiffHor	Horizontal diffuse irradiation
T_Amb	Ambient Temperature
GlobInc	Global incident in coll. plane
GlobEff	Effective Global, corr. for IAM and shadings
EArray	Effective energy at the output of the array
E_Grid	Energy injected into grid
PR	Performance Ratio

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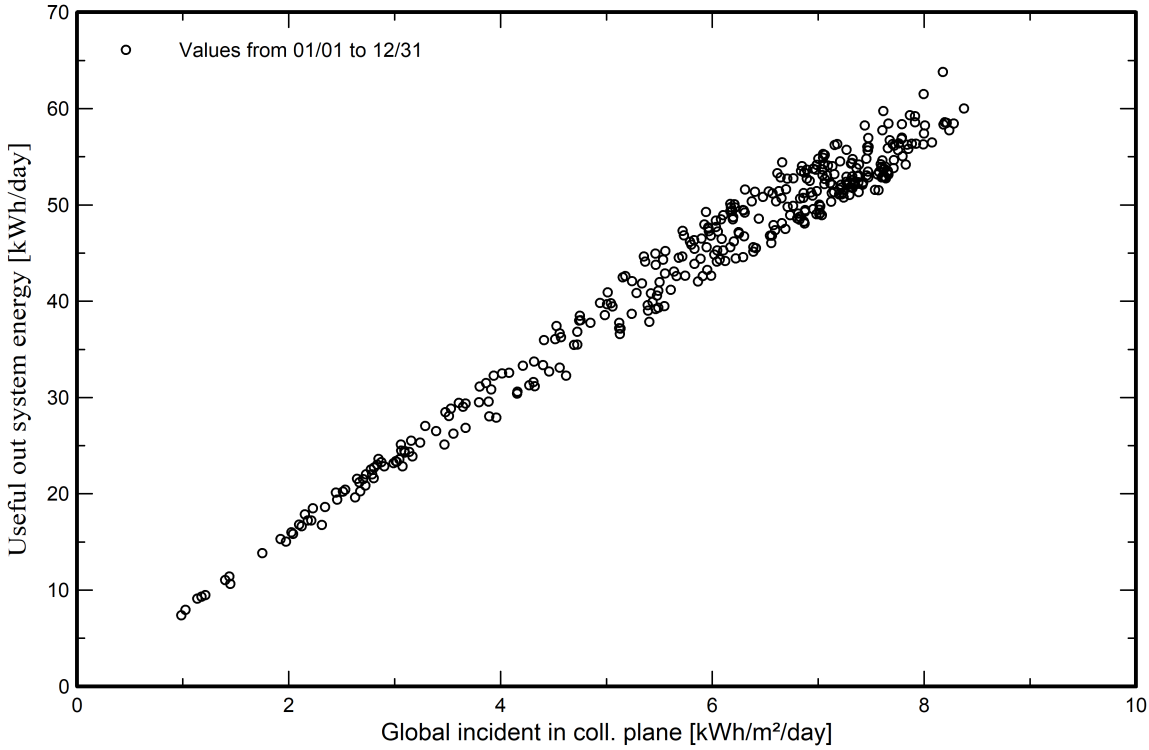
Loss diagram



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Predef. graphs

Daily Input/Output diagram



System Output Power Distribution

